SYNERGY TELECOM PVT LTD

10W Wideband Power Amplifier 2GHz~18GHz

Features

- Wideband Solid State Power Amplifier
- Gain: 52dB Typical
- Psat: 40dBm Typical
- Supply Voltage: +28V

Typical Applications

- Wireless Infrastructure
- 5G communication
- Test and measurement Instrument

RF Microwave & VSAT
Fiber Optics

Parameter		Min.	Typ.	Max.	Units
Frequency Range		2		18	GHz
Gain		45	52		dB
Gain Flatness			±2.0		dB
Gain Variation Over Temperature (-40℃~+70℃)			±3.0		dB
Input Return Loss			10		dB
Saturated Output Power (Psat)		38	40		dBm
Harmonics@37dBm			-20		dBc
Isolation S12			-60		dB
Supply Current (Vcc=+28V)			2.3	3	A
Power-Added Efficiency			10		%
Turn On/Off Speed (Gate Disable)	ON		100		us
	OFF		600		us
Turn On/Off Speed (AMP Disable)	ON		250		us
	OFF		20		us
Turn On/Off Speed (Switch Disable)	ON		100		ns
	OFF		100		ns

Weight	Net	39 Max. ounce	Impedance	50ohms
	Including Heat Sink	130.6 Max. ounce		
Input / Output Connectors		SMA-Female	Material	Copper
Finish		Nickel Plated	Package Sealing	Epoxy Sealed (Standard)
				Hermetically Sealed (Optional)

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PH:- 011-28533349, MOB:- 91-7217885948, FAX:- 011-28533349
EMAIL:- INFO@SYNERGYTPL.COM, WEBSITE:- WWW.RFCONNECTOR.IN

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Absolute Maximum Ratings

Operating Voltage	+30V
RF Input Power	Psat – Large Signal Gain

Note: Maximum RF input power is set to assure safety of amplifier. Input power may be increased at own risk to achieve full power of amplifier. Please reference gain and power curves.

Biasing Up Procedure

Step 1	Connect Ground Pin
Step 2	Connect input and output with 50 Ohm source/load. (in band VSWR10dB return loss)
Step 3	Connect +28V

Power OFF Procedure

Step 1	Turn off +28V biasing
Step 2	Remove RF connection
Step 3	Remove Ground

Environmental Specifications

Operational Temperature	-40ºC~+70ºC(Case Temperature)
Storage Temperature	-50ºC~+105ºC
Altitude	30,000 ft. (Epoxy Sealed Controlled environment)
	60,000 ft. 1.0psi min (Hermetically Sealed Un-controlled environment) (Optional)
Vibration	25g RMS (15 degrees 2KHz) endurance, 1 hour per axis
Humidity	100% RH at 35ºc, 95%RH at 40ºc
Shock	20G for 11msec half sine wave,3 axis both directions

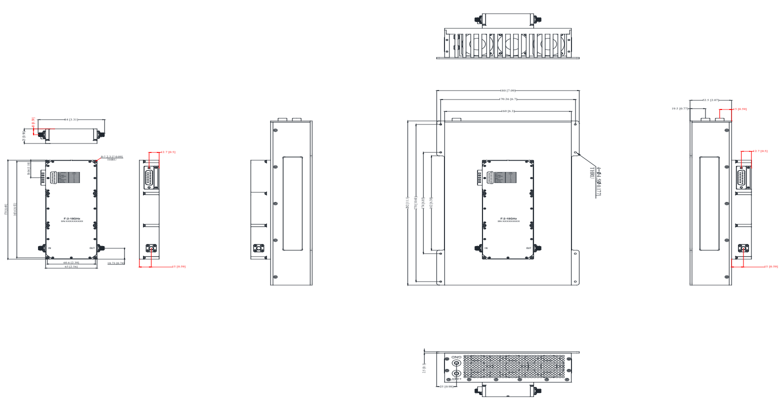
Note: The operating temperature for the unit is specified at the package base. It is the user’s responsibility to ensure the part is in an environment capable of maintaining the temperature within the specified limits

Outline Drawing:

All Dimensions in mm (inches)

Tolerance:±0.2(±0.008)(Excl Heat Sink)

Heat Sink required during operation(Sold Separately)



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Packing List

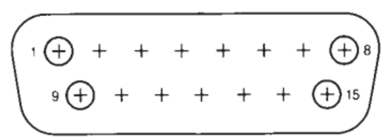
ID	Description	QTY
1	Fig a. Fan adapter	1
2	Fig b. DB15 cable (51321000015)	1

Fig a.

Fig b.

Protection Connector Table

Male D-Sub is on the housing
The mating female part number: 172-E15-203R001



Pin #	Name	Function	Initial State	Description	Applied
1,2,9,10	VDD	Power Supply	+28V	+28V DC is supply Voltage	Yes
3,11	GND	Ground	GND	Ground	Yes
4	+5V_USER	Power Supply	+5V	+5V DC is supplied for reference(200mA)	Yes
5	TEMP_ SIGNAL	Indicator	Voltage	Display temperature signal	Yes
6	GATE_OFF	Control	LOW	Applying logic HIGH disables gates of amplifiers	Yes
7	AMP_OFF	Control	LOW	Applying logic HIGH disables Positive Supply Voltage of amplifiers	Yes
8	RESET	Control	HIGH	Resets PA when logic LOW is applied and released (Internally Pulled-High +5V)	Yes
12	RF_Switch _OFF	Control	HIGH	Applying logic LOW disconnect RF signal of amplifiers	Yes
13	RF IN Over	Indicator	LOW	Pin will be latched to logic HIGH when input signal is over limit	Yes
14	Temp Over	Indicator	LOW	Pin will be latched to logic HIGH when drive over Temperature	Yes
15	Current Over	Indicator	LOW	Pin will be latched to logic HIGH when Current Limit is reached	Yes

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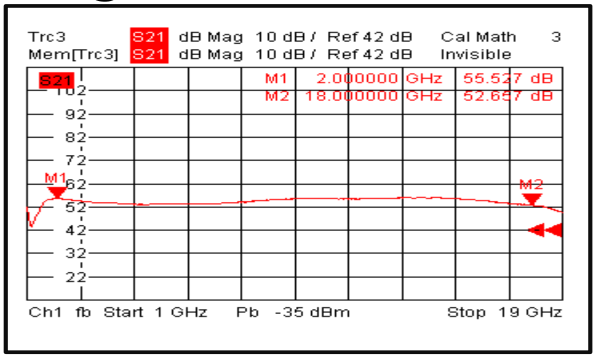
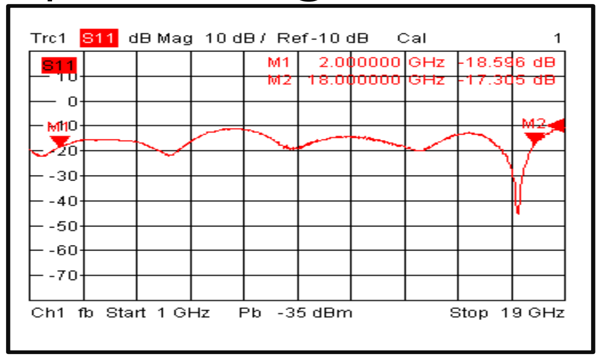
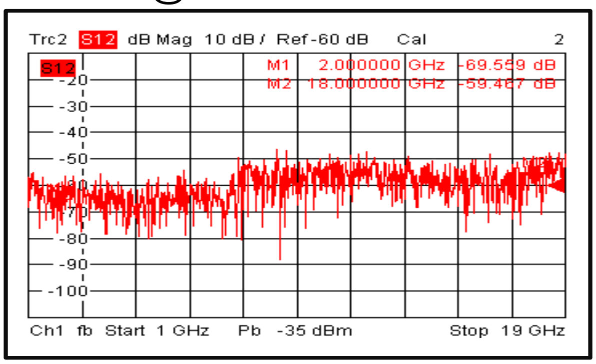
Gain @ +25℃



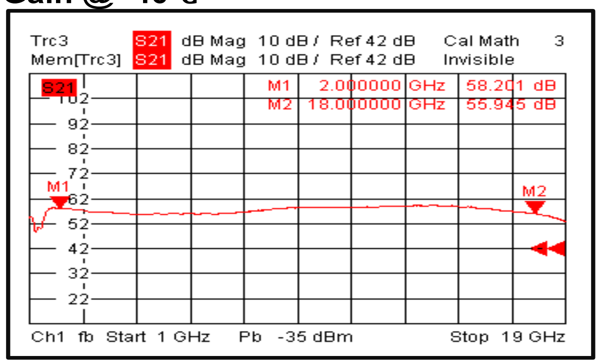
Input Return Loss@+25℃



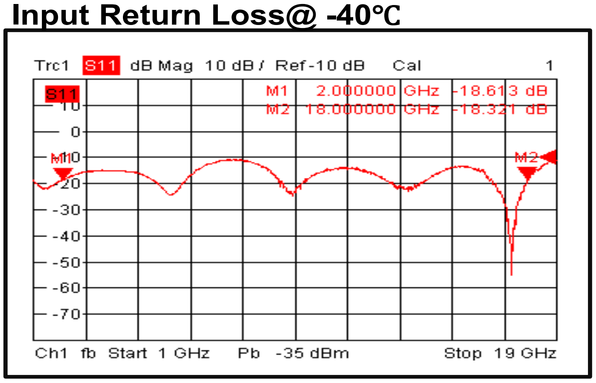
Isolation @ +25℃



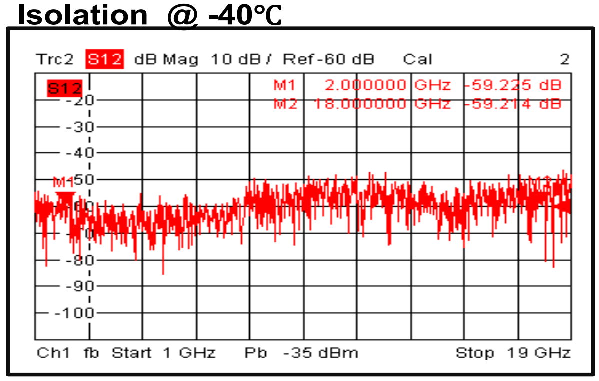
Gain @ -40℃



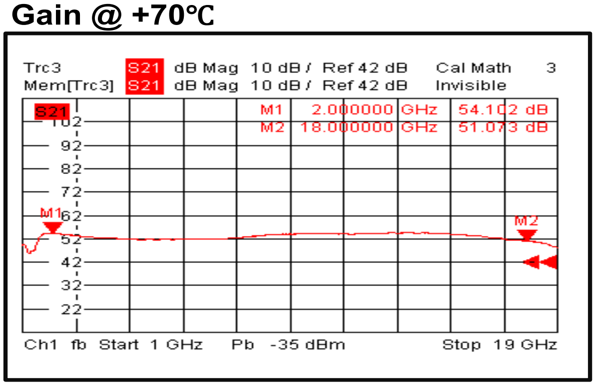
Input Return Loss@ -40℃



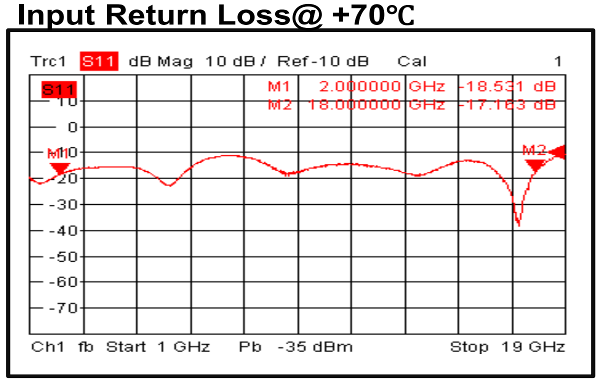
Isolation @ -40℃



Gain @ +70℃



Input Return Loss@ +70℃



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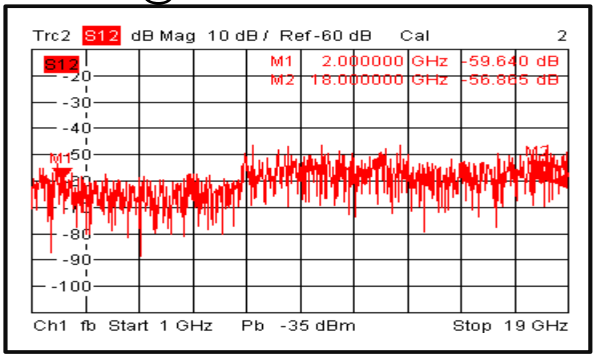
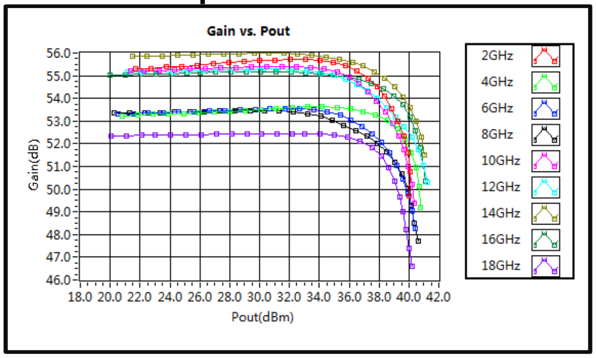
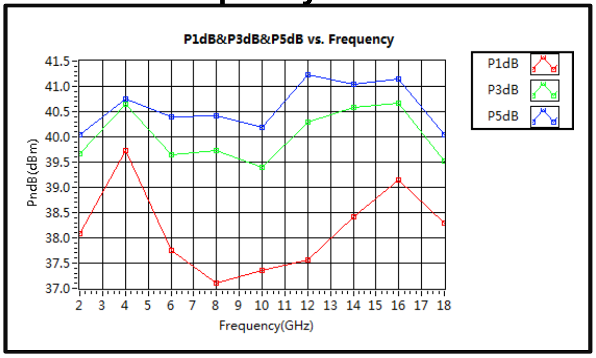
Isolation@+70℃



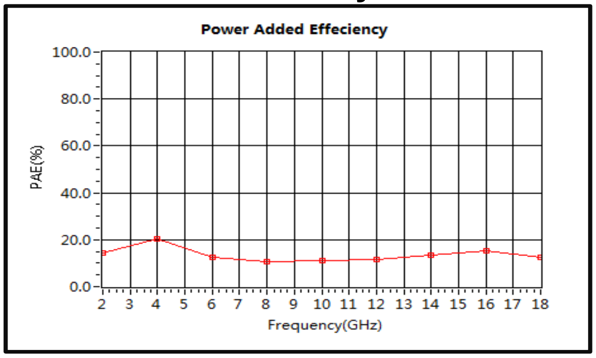
Gain vs. Output Power



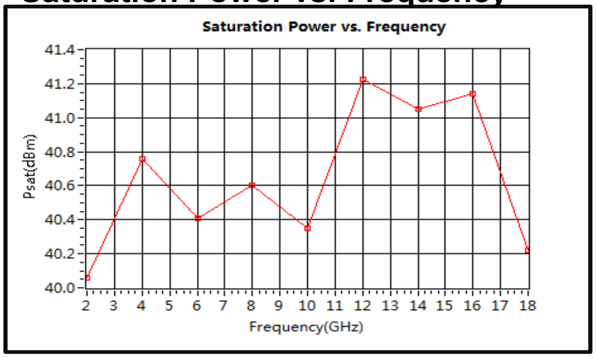
PndB vs. Frequency



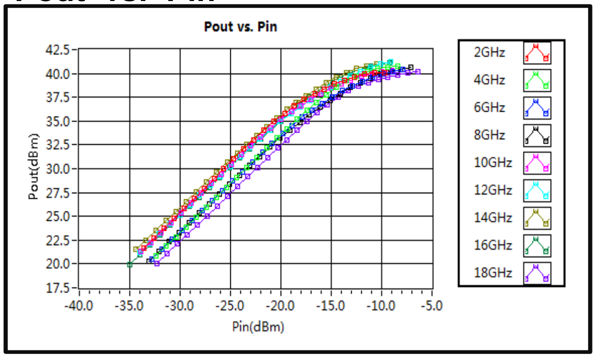
Power Added Efficiency



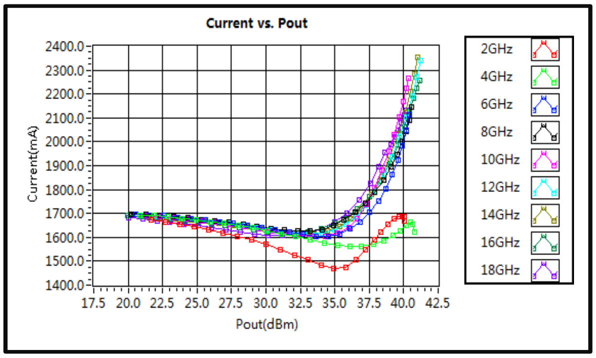
Saturation Power vs. Frequency



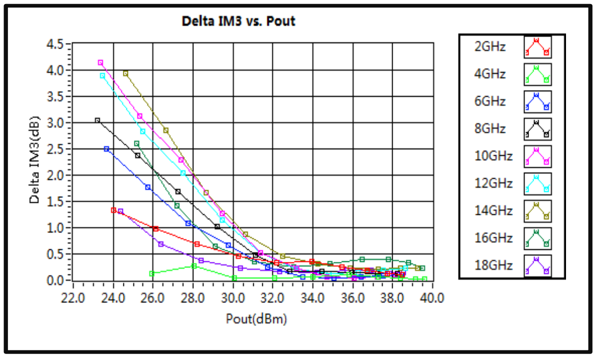
Pout vs. Pin



Current vs. Pout



Delta IM3 vs. Pout



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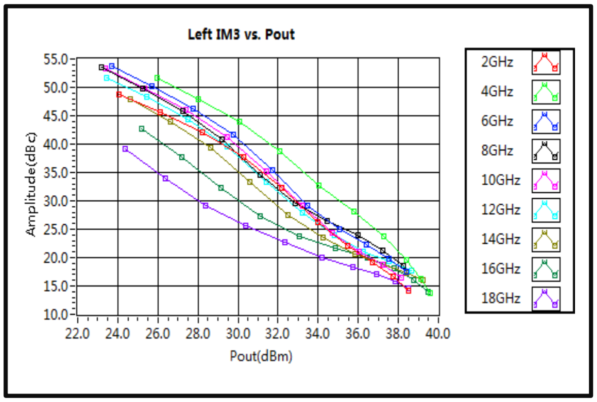
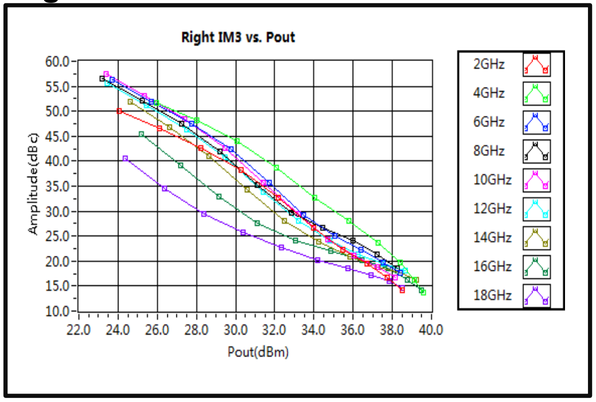
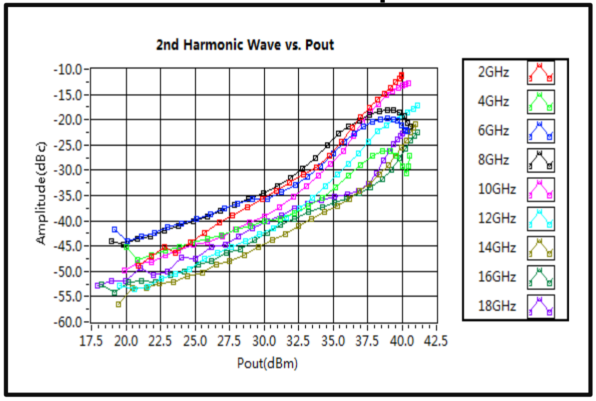
Left IM3 vs. Pout



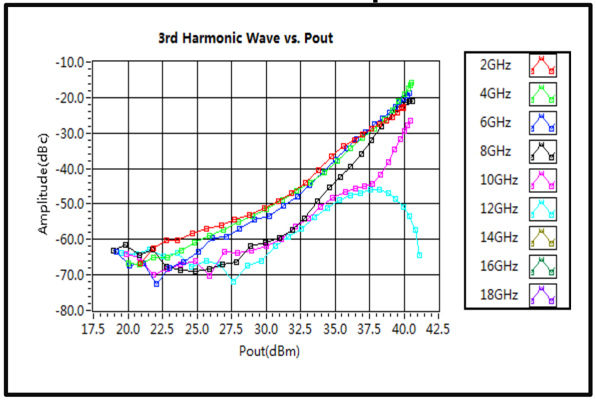
Right IM3 vs. Pout



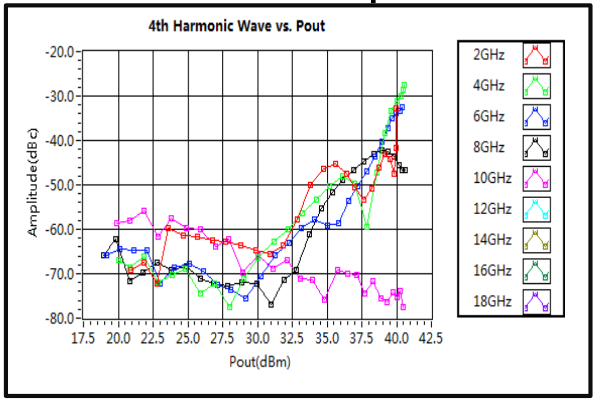
2nd Harmonic Wave Output Power



3rd Harmonic Wave Output Power



4th Harmonic Wave Output Power



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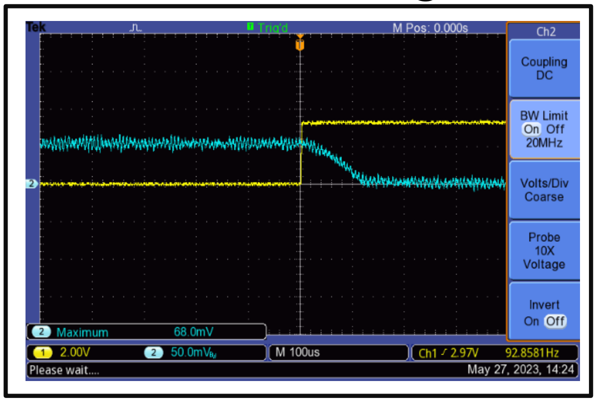
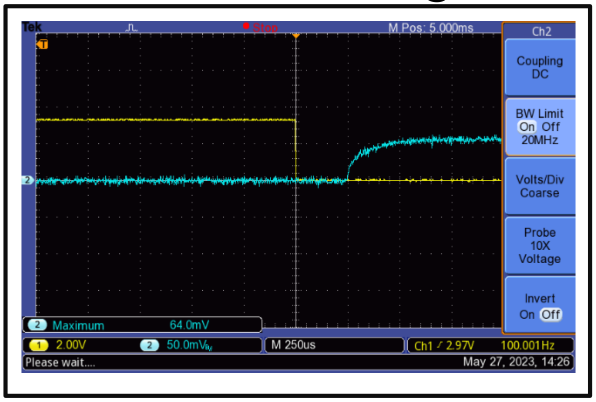
The Gate-Enable Time is 100 us @+25℃

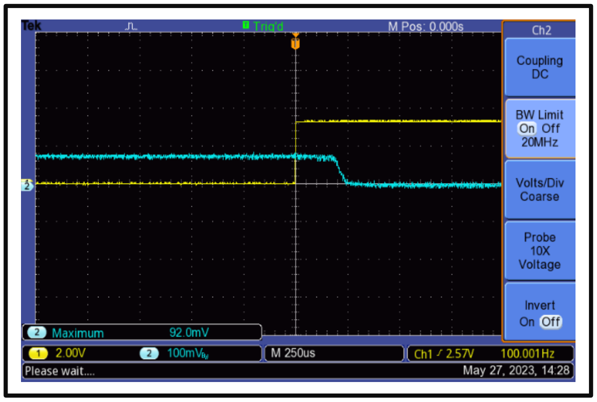


The Gate-Disable Time is 600 us @+25℃

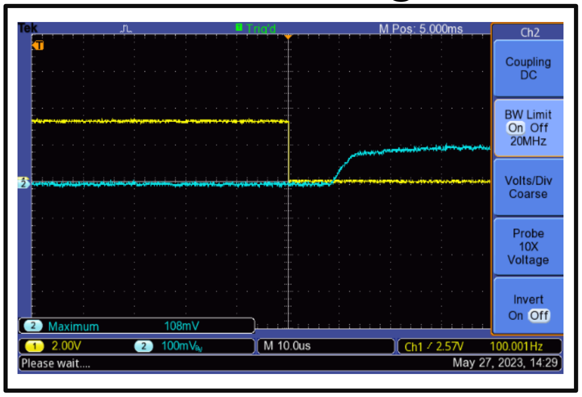


The gate control port：D-sub 15 PIN #6 (GATE_OFF).
The yellow curve is the gate control signal, the blue curve is RF output envelope.

The AMP**-Enable Time is 250 us @+25℃**

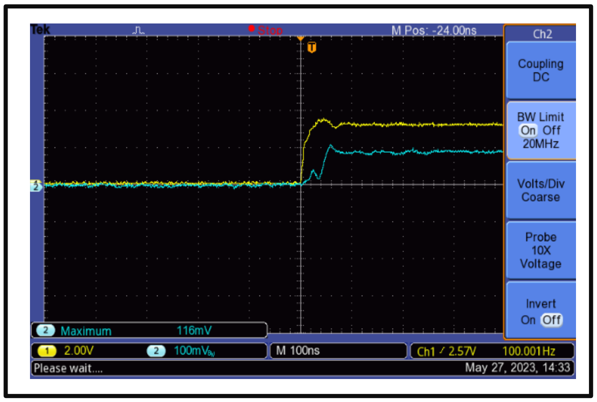


The AMP**-Disable Time is 20 us @+25℃**

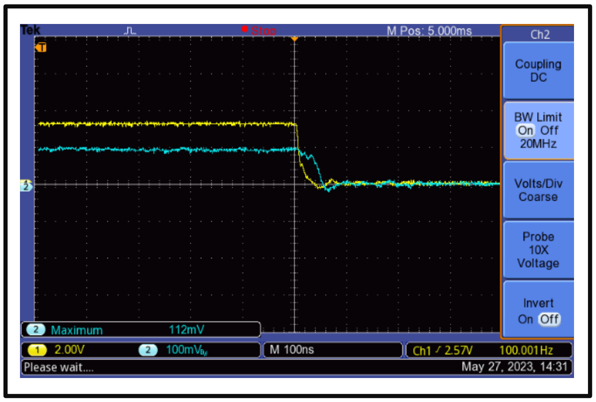


The AMP control port：D-sub 15 PIN #7(AMP _OFF).
The yellow curve is the AMP control signal, the blue curve is RF output envelope.

The Switching Rise Time is 100 ns @+25℃



The Switching Fall Time is 100 ns @+25℃



Switch control port: D-sub 15 PIN #12(RF_Switch_Off) .
The yellow curve is the switch control signal, the blue curve is RF output envelope.

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